

Detector Ranking is Not Architecture-Invariant under Medical Covariate Shift

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Abstract

Post-hoc out-of-distribution (OOD) detectors are almost always ranked on a single backbone. **Holding a medical covariate split fixed, detector ranking is governed more by representation family than by detector identity.** On Camelyon17 hospital-2 (WILDS), Kendall's $W = 0.694$ across eight CNN families and seven scores (seed 42). MSP AUROC spans 0.515 (ResNet-50) to 0.883 (DenseNet-121)—a 0.37 swing, as large as Mahalanobis—MSP on that ResNet. Copying ResNet-50's best logit score (ReAct) onto ConvNeXt costs 0.185 AUROC versus ConvNeXt's own best logit. Selecting the AUROC-winning backbone for OOD triage yields coverage@risk 10% = 0.078; selecting by coverage itself yields 0.904. Feature-space scores occupy ranks 1–2 on every official backbone. Skin ISIC→PAD does not reproduce the MSP jump ($W = 0.791$). AUROC versus coverage@risk disagreement flips sign between Camelyon and MIDOG. A pre-registered ViT-B/16 MSP of 0.884 met a locked threshold of 0.6947. The protocol: do not copy an OOD ranking across backbones, and do not pick a backbone for triage from an AUROC table.

1. Introduction

We test one hypothesis: *under a fixed medical covariate shift, post-hoc detector ranking is an architecture property.* On Camelyon17 hospital-2, ResNet-50 MSP AUROC is 0.515 while DenseNet-121 is 0.883 (Fig. 1a). A lab that copies an OpenMIBOOD-style ResNet-50 recipe and a lab that uses DenseNet are not ranking the same detectors—and if they then pick a backbone from that AUROC table for triage, they can throw away 0.83 of OOD coverage at a 10% risk budget (Fig. 1b).

That hypothesis is *not* that ranking was previously assumed architecture-invariant. On natural-image semantic OOD, ranking already moves with architecture [2, 3, 23, 31]. OpenMIBOOD [8] already showed that ImageNet method ranking does not transfer to medical data and that

feature-space scores outperform logit scores under a frozen classifier per domain. The orthogonal question is: **holding one medical covariate-shift dataset fixed, how much does ranking move when only the backbone changes?**

The rest of the paper is that test, in order. Camelyon discovers the shift (Kendall $W = 0.694$, MSP jump). Five seeds verify the DenseNet pole (5/5). Matched-recipe training asks whether the pole is architecture or optimizer. Skin falsifies a universal MSP-jump law, while feature ranks stay 1–2. MIDOG is the deployment readout: the *sign* of AUROC versus coverage@risk disagreement flips. A pre-registered ViT-B/16 tests whether the Camelyon pole extends past CNNs. The contribution is a protocol, not a detector.

Contributions.

- Camelyon discovers architecture-governed ranking.** Official Kendall $W = 0.694$ ($n=8$, 7 methods, seed 42). MSP AUROC span 0.515–0.883 matches the Maha–MSP gap on ResNet-50. Copying ResNet-50's winning logit (ReAct) onto ConvNeXt costs 0.185 AUROC. Feature-space scores stay in ranks 1–2. The seed-robust MSP jump is DenseNet-121 (0.265 ± 0.065 , 5/5); ConvNeXt and EffV2-S@224 are 4/5, MobileNet/RegNet 3/5, EffB3 2/5.
- Skin falsifies universality.** ISIC→PAD ($n=8$, $W=0.791$): Maha rank-1 and k NN rank-2 on all eight; MSP stays in a narrow band. Not a second Camelyon jump.
- Deployment cost is unpredictable from AUROC.** The sign of AUROC versus coverage@risk 10% disagreement flips between Camelyon and MIDOG [6, 26]. On Camelyon $n=8$, copying the AUROC ranking costs 0.826 coverage at a 10% risk budget.
- A locked prediction.** Before training: ViT-B/16 Camelyon MSP ≥ 0.6947 . Observed 0.884. Held out of the 8-CNN W .

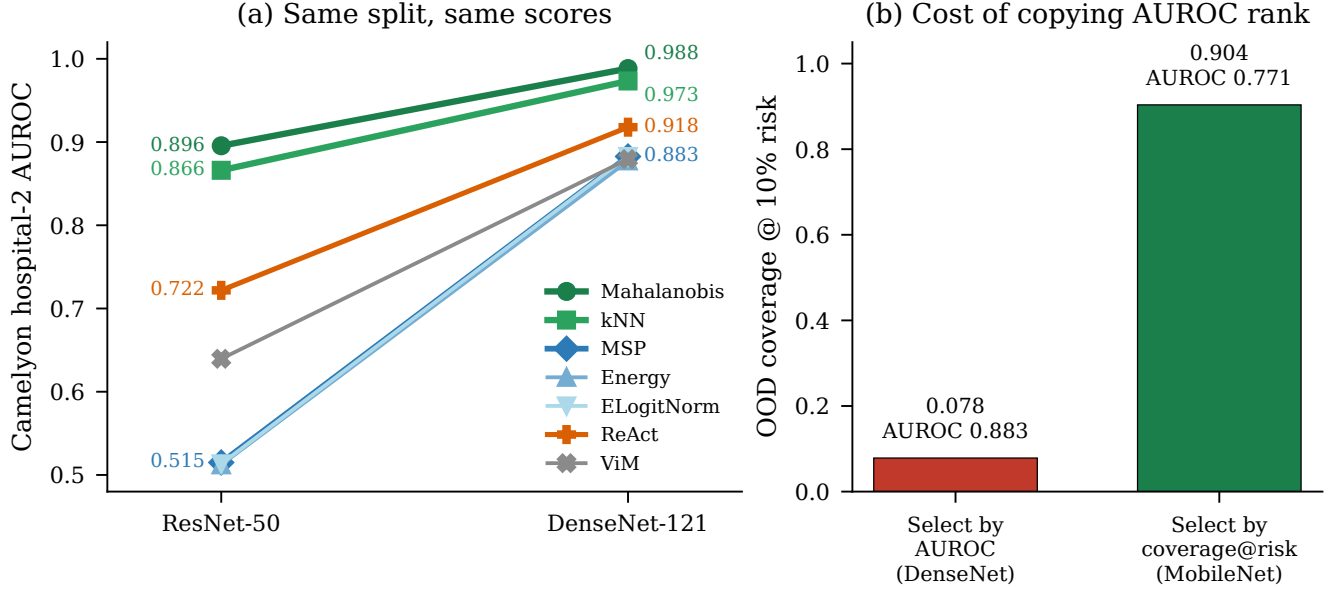


Figure 1. **The claim in one figure.** (a) Same Camelyon hospital-2 split, same seven scores: switching ResNet-50→DenseNet-121 lifts logit AUROC by ~ 0.37 while Mahalanobis and k NN stay on top. (b) If a practitioner copies the AUROC ranking and deploys DenseNet for OOD triage, coverage@risk 10% is 0.078; measuring coverage itself selects MobileNet at 0.904.

Work	#arch.	Shift	Rank stat.	Utility
OpenOOD v1 [29]	1/set	semantic	—	AUROC
OpenOOD v1.5 [31]	3	semantic	qualitative	AUROC
Szyc <i>et al.</i> [23]	3 var.	CIFAR sem.	rank Δ	AUROC
Datko <i>et al.</i> [3]	per line	natural	claimed stable	AUROC
OpenMIBOOD [8]	1/domain	medical	—	AUROC
Ours	8+ViT	covar.	Kendall W	AUROC+cov.

Table 1. Closest ranking-vs-backbone papers. Architecture-sensitive ranking exists on natural-image semantic OOD; the increment here is a held-fixed medical covariate split, zoo-scale W , and operational coverage.

2. Related work

OpenMIBOOD [8] ranks 24 post-hoc detectors with a **single frozen classifier per medical domain**. It shows that ImageNet method ranking does not transfer to medical data and that feature-space scores outperform logit scores under that protocol. Table 1 locates the increment: we hold the dataset fixed and change the backbone.

OpenOOD v1 [29] fixes architecture on purpose. OpenOOD v1.5 [31] compares three ImageNet models on semantic near/far OOD and reports that “different post-processor may favor different architecture,” without Kendall’s W and without a covariate hospital split. Szyc *et al.* [23] show rankings change across three ResNet CIFAR implementation variants. Datko *et al.* [3] claim an architecture-specific winner that transfers across natural-image OOD—the opposite of a single-split concordance statistic. Claros Olivares and Brockmeier [2] rank confi-

dence scores with AURC/AUGRC on CIFAR / TinyImageNet; they do not report a sign-flip of AUROC versus coverage@risk 10% across medical covariate domains.

OoD-Bench [30] is OOD generalization, not detection. WILDS [14] supplies Camelyon17 and iWildCam; iWildCam is a negative bound, not a third main domain.

3. Setup

3.1. Shifts

Camelyon17 WILDS [14]: hospitals 0, 3, 4 as ID and hospital 2 as OOD; founding cell, official W , $n=8$. Skin ISIC→PAD [19]: feature rank-stability without an MSP jump. MIDOG [8]: OpenMIBOOD 1a vs. 1b+1c, coverage sign-flip; official W is $n=3$. We build on OpenMIBOOD’s public MIDOG split and ResNet-50 checkpoint as an external reference. Public R50 AUROC ~ 0.59 is *theirs*; in-house R50 MSP 0.512 is *ours*. We never mix the two.

3.2. Backbones

Eight CNN-family models, ImageNet initialization, 10-epoch fine-tune, seed 42, train-frac 1.0 on Camelyon: ResNet-18/50 [9], DenseNet-121 [12], ConvNeXt-Tiny [18], MobileNetV3-Large [11], RegNetY-3.2GF [20], EfficientNet-B3 [24] at native 300, and EfficientNetV2-S [25] at 224 (native 384 dropped). Standard fine-tune recipes (Adam / AdamW / SGD) follow each family’s common practice and are *not* forced-identical Adam. That con-found is measured in Sec. 4.3. Only resolution was isolated

in a controlled pair (EffB3 @224 vs. @300; EffV2-S @224 vs. @384). ViT-B/16 [4] @224 is held out of W .

3.3. Scores and statistics

Seven post-hoc scores: MSP [10], Energy [17], energy of logit-normed logits (ELogitNorm), ViM [27], ReAct [21], Mahalanobis with Ledoit-Wolf covariance [15, 16], and k NN [22]. Primary concordance: **Kendall's W per domain, never pooled**. Jump: $\text{MSP} - \text{mean}(\text{R18}, \text{R50}) \geq 0.15$. Feature drag: Maha or k NN on a new backbone below $\text{median}(\text{R18}, \text{R50}, \text{EffB3}) - 0.10$. Official Camelyon ResNet mean $\text{MSP} = 0.544654$; jump iff $\text{MSP} \geq 0.695$ on that split.

Coverage@risk 10% is OOD-triage: rank OOD samples by MSP (higher = more ID-like) and take the largest coverage whose classifier risk is $\leq 10\%$. ID-selective Camelyon is vacuous (ID-val acc $\approx 99.5\%$) and is not cited.

We tested whether the Camelyon MSP pattern is an artifact of input resolution, temperature scaling [7], ID-only detector *selection*, a CE+SupCon recipe [13], or train-time HED stain jitter. None is a clean account (Sec. 5 and the supplement).

4. Results

4.1. Camelyon-17 discovers the ranking shift

Official Kendall $W = 0.694$ ($n=8$, 7 methods). This is the only Camelyon number called “official W .” Leave-one-backbone recomputation of W ranges 0.676–0.738; that is *structural* sensitivity of the concordance statistic, not a 95% CI (supplement).

Feature methods occupy ranks 1–2 on every official backbone (Maha #1 on 6/8; k NN #1 on EffB3 and MobileNet). No backbone drags Maha/ k NN past the 0.10 feature-drag threshold. On Table 2, MSP spans 0.515–0.883; Maha 0.870–0.988; k NN 0.856–0.973. Incomplete concordance ($W=0.694$) is the logit ranks moving (Fig. 2).

Table 3 reports the full five-seed MSP matrix: seeds verify the DenseNet pole. DenseNet-121 is the confirmed seed-robust jump (5/5). ConvNeXt-Tiny and EfficientNetV2-S@224 are not seed-42 flukes (4/5) and are not promoted to 5/5. MobileNetV3 and RegNetY are 3/5; EfficientNet-B3 is 2/5 (its seed-42 jump +0.258 is a favorable draw relative to 0.142 ± 0.125). The seed-42 matrix still shows a jump without feature drag. The claim is narrower than “6/6 non-ResNet, confirmed.”

Copying a logit ranking is not free. ResNet-50's best logit is ReAct (0.722); on ConvNeXt the same score is 0.692 against Energy 0.877—a 0.185 AUROC transfer cost. Feature-space transfer regret is ≤ 0.05 because Maha/ k NN stay in ranks 1–2. Architecture moves the logit leaderboard; it does not unseat the feature default.

4.2. Skin falsifies a universal MSP jump

Official $W = 0.791$ ($n=8$). Maha is rank-1 and k NN rank-2 on **all eight** (Fig. 2). MSP sits in a narrow band (0.68–0.74). There is no Camelyon-style ResNet-vs-rest MSP jump. EffV2-S@224 MSP is 0.737; the native-384 cell that put k NN at rank 7 (MSP 0.823) is a resolution artifact and is not in official W . Skin is rank-stability of feature scores under a zoo: a replication of OpenMIBOOD's feature>logit pattern *across architectures*, not an MSP-jump.

4.3. Optimizer challenges a clean architecture reading

Resolution is not the Camelyon DenseNet / seed-42 pattern. EffB3 Camelyon: 0.803 @300 $\rightarrow$ 0.787 @224 (the jump versus the official ResNet mean survives). EffV2-S Camelyon: 0.726 @384 $\rightarrow$ **0.778** @224 (the official cell; the jump is not a 384 artifact). On skin, the EffV2-S @384 k NN collapse *is* a resolution artifact and disappears at 224. These two facts are not a slogan that “resolution never matters.”

M1 — jumpers trained with ResNet Adam ($\text{lr}=10^{-4}$, $\text{wd}=10^{-4}$, batch 64). The four official zoo members whose standard recipe was not that Adam still jump versus the frozen official ResNet mean 0.545: ConvNeXt 0.820 (+0.275), MobileNetV3 0.737 (+0.193), RegNet 0.805 (+0.261), EffV2-S@224 0.873 (+0.328); all with id-val acc ≥ 0.994 . 4/4 clean hits. DenseNet and EffB3 already shared ResNet-Adam in the official zoo. Recipe mismatch does *not* explain the seed-42 jumps of those four members. M1 is seed 42 only; the five-seed MSP matrix is Table 3.

M2 — ResNets trained with ConvNeXt AdamW ($\text{wd}=0.05$). ResNet-18 MSP **0.773** (enters the 0.695 band; id-val 0.9957). ResNet-50 MSP 0.581 (stays out; id-val 0.9958). ResNet *can* jump under a non-default recipe. Architecture is not a clean variable, independent of optimizer. These two results point in opposite directions; we report both.

4.4. ViT-B/16: a locked prediction

Locked *before* training, from the seed-42 zoo rule that non-ResNet-like cells jumped: $\text{MSP} \geq 0.6947$. Linear depth/parameter/receptive-field predictors are invalid on this $n=8$ and are not quoted. Observed MSP **0.884**, Maha 0.958, k NN 0.941. The prediction holds. Not a ninth W column.

4.5. Coverage@risk: the deployment cost

Protocol: OOD-triage coverage@risk 10% with MSP, not ID-selective coverage.

On Camelyon hospital-2, the locked $n=4$ utility table has MSP AUROC order EffB3@300 (0.803) > EffB3@224

Backbone	input	Maha (rank)	kNN (rank)	MSP	seed-42 jump
ResNet-18	224	0.956 (1)	0.936 (2)	0.574	no
ResNet-50	224	0.896 (1)	0.866 (2)	0.515	no
DenseNet-121	224	0.988 (1)	0.973 (2)	0.883	yes
ConvNeXt-Tiny	224	0.928 (1)	0.916 (2)	0.846	yes
MobileNetV3-L	224	0.870 (2)	0.904 (1)	0.771	yes
RegNetY-3.2GF	224	0.964 (1)	0.956 (2)	0.784	yes
EfficientNet-B3	300	0.885 (2)	0.934 (1)	0.803	yes
EfficientNetV2-S	224	0.880 (1)	0.856 (2)	0.778	yes

Table 2. Camelyon hospital-2 AUROC, **seed 42** (official W). Jump iff $\text{MSP} \geq 0.6947$ vs. frozen ResNet mean 0.544654. MSP range is ResNet-50→DenseNet.

Backbone	42	43	44	45	46	jump mean $\pm$ sd	≥ 0.15
ResNet-18	0.574	0.712	0.601	0.524	0.581	—	—
ResNet-50	0.515	0.621	0.619	0.515	0.661	—	—
ResNet mean	0.545	0.667	0.610	0.520	0.621	—	—
DenseNet-121	0.883	0.865	0.868	0.845	0.826	0.265 $\pm$ 0.065	5/5
ConvNeXt-Tiny	0.846	0.799	0.803	0.764	0.785	0.207 $\pm$ 0.067	4/5
MobileNetV3-L	0.771	0.728	0.815	0.780	0.745	0.175 $\pm$ 0.081	3/5
RegNetY-3.2GF	0.784	0.778	0.836	0.830	0.724	0.198 $\pm$ 0.089	3/5
EfficientNet-B3	0.803	0.731	0.643	0.818	0.678	0.142 $\pm$ 0.125	2/5
EfficientNetV2-S@224	0.778	0.744	0.808	0.846	0.835	0.210 $\pm$ 0.089	4/5

Table 3. Same eight Camelyon hospital-2 MSP cells on all five seeds {42, 43, 44, 45, 46}. Jump = MSP minus that seed’s ResNet mean; bar 0.15. The last column is how many of those five seeds clear the bar. Official W is seed 42 only.

(0.787) > ResNet-18 (0.574) > ResNet-50 (0.515), but coverage@risk 10% order ResNet-50 (0.149) > ResNet-18 (0.145) > EffB3@300 (0.107) > EffB3@224 (0.013). The AUROC winner is not the coverage winner (Fig. 3).

On MIDOG, official $n=3$ (in-house): MSP AUROC ResNet-50 0.512, EffB3 0.486, ResNet-18 0.438 (gaps are small). Coverage@risk 10%: EffB3 0.712, ResNet-18 0.442, ResNet-50 0.166. ResNet-50 is AUROC-highest and coverage-lowest. The permutation is *not* the Camelyon permutation.

AURC/AUGRC already assume that AUROC and coverage-based ranking can disagree [6, 26]. The increment is the *sign* of that disagreement flipping between two medical covariate domains under the same protocol, with no rule for which backbone wins. On the Camelyon $n=8$ zoo, copying the AUROC ranking (DenseNet, coverage 0.078) instead of measuring coverage (MobileNet, 0.904) costs 0.826 OOD mass at a 10% risk budget (Fig. 1b). Measure coverage@risk on the deployment domain.

4.6. Frozen foundation models

Camelyon linear-probe MSP does not invert: GigaPath [28] 0.664, Phikon [5] 0.605, UNI [1] 0.599, all >0.5. MIDOG UNI / GigaPath MSP 0.445 / 0.470 is a MIDOG-only footnote. FM Maha ~ 0.995 –1.0 is discussion support that feature>logit can survive a training-paradigm change, not a

pillar.

5. Failed alternatives

Each of the following was a precommitted test. None is hidden.

Resolution (from Sec. 4.3). EffB3@224 MSP 0.787; EffV2-S@224 MSP 0.778. Not a pixel-size explanation of Table 2.

Temperature T^* . Correlation of T^* with the MSP jump: $r=0.268$, $p=0.52$. DenseNet–ResNet-18 MSP gap 0.308→0.308 after T^* . Calibration/temperature is not the mechanism.

ID-only Maha/MSP gate. Selecting a score from ID-only statistics is a *selector*, not a detector. On Camelyon it picks Maha on 2/8 backbones; mean AUROC of the gated score 0.837 versus always-Maha 0.940. The bar was matching Maha and ViM. It does not.

CE+SupCon ($\lambda=0.1$, $\tau=0.07$), ResNet-18 + DenseNet. Gap 0.308→0.238 (23%; precommit bar was 50% / ≤ 0.154). ResNet-18 MSP 0.574→0.515 (worse; this is the SupCon cell, not official ResNet-50 MSP 0.515). Part of the shrink is the low model degrading. Dirty; not a recipe.

HED stain jitter, an in-house MIDOG $n=8$ zoo, and a shift-type extract (covariate / task / far-OOD) are in the sup-

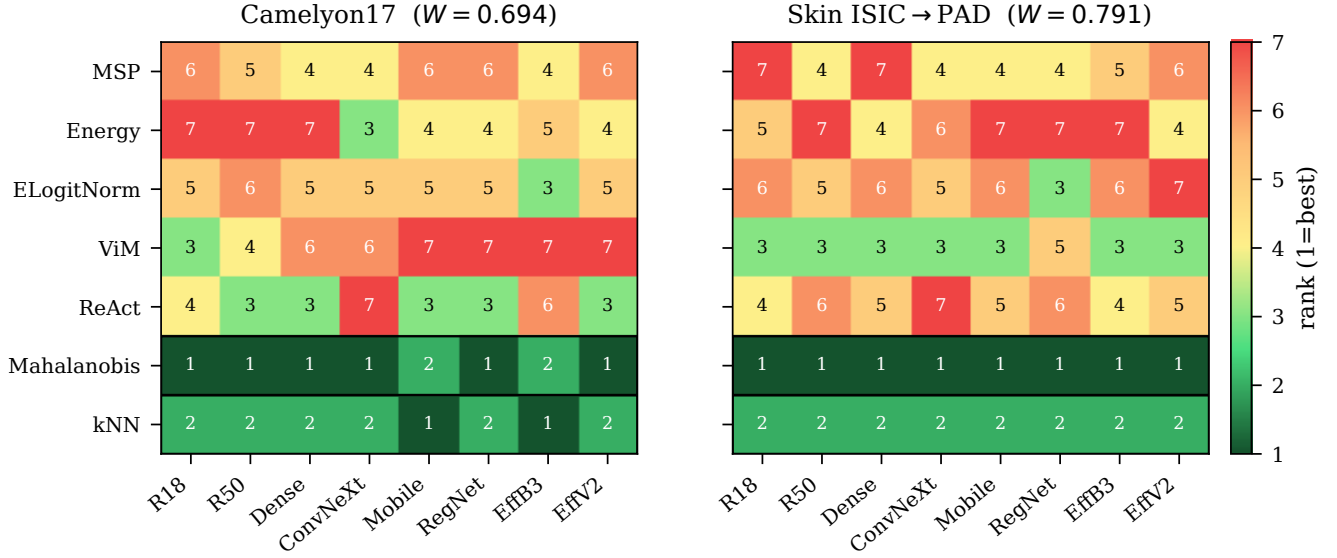


Figure 2. **Method rank by backbone** (1=best AUROC). Feature-space scores occupy ranks 1–2 on every official backbone in both domains. Logit ranks move on Camelyon ($W=0.694$) and much less on skin ($W=0.791$), where Maha is rank-1 on 8/8.

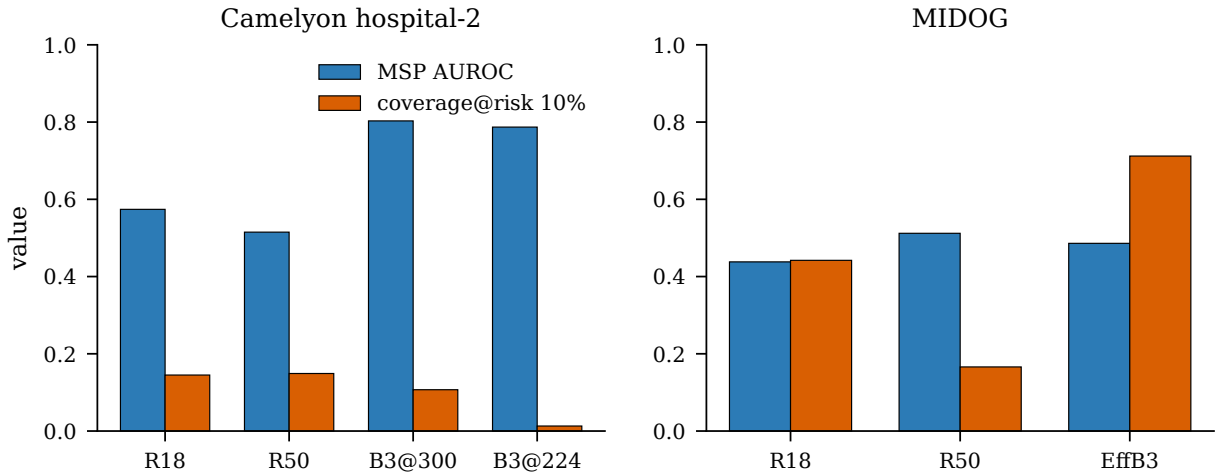


Figure 3. **AUROC vs. coverage@risk 10%**. On Camelyon the AUROC-high EfficientNets have the *lowest* triage coverage; on MIDOG the AUROC-high ResNet-50 is coverage-lowest. The sign of the disagreement flips.

plement. Joint readout: no shift-type law, no stain mitigation, no Camelyon-style MSP jump on MIDOG.

6. Discussion and limitations

The hypothesis holds on Camelyon and fails as a cross-medical MSP-jump law. Representation family—ResNet versus the rest, feature space versus logits—moves ranking and utility more than swapping MSP for Energy or ReAct on a frozen classifier. Feature-space scores remain the stable default when the shift looks like Camelyon *and* when it does not (skin). M1 preserves the seed-42 jumps under ResNet-Adam; M2 lets ResNet-18 enter the MSP band un-

der ConvNeXt-AdamW: architecture is the dominant axis, not a clean one. ViT-B/16 was a locked prediction, not a post-hoc ninth CNN.

The practical cost is not abstract. Copying an AUROC ranking on Camelyon discards 0.83 of OOD coverage at a 10% risk budget. Copying ResNet-50’s logit winner onto ConvNeXt costs 0.185 AUROC. Those are deployment errors from transferring a ranking, not from picking a weak detector in isolation.

Limitations. (1) **iWildCam bound.** Locked 3-backbone cell: DenseNet MSP 0.575 vs. ResNet mean 0.604, jump -0.029 . Extra five CNNs vs. that domain’s ResNet mean:

ConvNeXt 0.601, MobileNet 0.558, RegNet 0.580, EffB3 0.577, EffV2-S 0.650; 0/5 jump. The Camelyon MSP-jump is absent on this non-medical covariate shift and may be biomedical-specific. iWildCam is not pooled with Camelyon W . (2) **No clean mitigation.** CE+SupCon is dirty (ResNet-18 MSP worse). HED stain jitter is dirty (DenseNet MSP $0.780 < 0.80$). The ID-only gate is a selector. (3) **MIDOG official W is $n=3$** ($W=0.857$); the in-house $n=8$ zoo shows no MSP jump and does not replace it. (4) W is per-domain; $0.694 / 0.791 / 0.857$ are not averaged. (5) No reader study: the operational claim stops at coverage@risk. Coverage@risk itself is a point estimate here; per-sample bootstrap CIs are left to the toolkit release (features are not re-extracted in this draft). (6) Architecture is confounded with its standard recipe. 5/5 seed-robustness is DenseNet only. Official W stays seed 42. Staining vs. scanner vs. tissue prep inside the hospital shift is not isolated.

Code for the architecture-zoo evaluation and coverage@risk will be released, crediting OpenMIBOOD for MIDOG imglits, the public ResNet-50 checkpoint, and the 1a / cs-ID / near-OOD protocol.

7. Conclusion

Detector ranking under medical covariate shift is governed more by representation family than by detector identity. On Camelyon that is a seed-robust DenseNet MSP pole, stable feature ranks, and a 0.83 coverage penalty for copying AUROC into triage. Skin keeps the feature ranks and drops the MSP jump. MIDOG flips the sign of the AUROC-coverage disagreement. A pre-registered ViT hit shows the Camelyon logit instability is not a CNN-zoo accident; iWildCam does not reproduce it. The protocol is the result: rank detectors on the backbone you will deploy, and pick that backbone from coverage@risk on the deployment shift.

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